

Team Name: Jexko-1Bit

Project Title: TaniCare AI – Kawan Petani, Penjaga Tanaman

Hackathon: MyAI Future Hackathon 2026 –

Track 1: Padi & Plates (Agrotech & Food Security)

Problem Malaysia's rice self-sufficiency ratio is only ~52% in 2026, far below the NAP 2.0 target of 80%. Smallholder farmers (padi, sawit, sayur, buah) lose 30–70% of yield every season due to late detection of crop diseases. Limited access to agronomists and lack of real-time tools worsen food import dependency and threaten farmer livelihoods.

Solution TaniCare AI is a smart, farmer-friendly mobile app that enables instant crop disease detection and advanced monitoring.

Key Features

- One-tap photo diagnosis using Google Gemini 1.5 Pro (accurate results in Bahasa Melayu)
- Advanced satellite analysis: NDVI + EVI + anomaly detection via Google Earth Engine
- Real-time weather integration (temperature, humidity, rain) for intelligent risk alerts
- Professional Analytics Dashboard with interactive NDVI trend charts and NDVI vs EVI bar comparison
- Dynamic support for all Malaysian states
- Scan history automatically saved to Google Cloud Firestore

Technology Stack

- Frontend: Flutter + Riverpod 2.0 (advanced state management)
- AI: Google Gemini 1.5 Pro
- Satellite: Google Earth Engine (Sentinel-2)
- Weather: Open-Meteo real-time API
- Database: Google Cloud Firestore (GCP)
- Charts: fl_chart
- Backend: Google Cloud Functions (Python)

Impact

- Expected to reduce crop yield losses by 30–50%
- Directly supports DKMN 2030 and NAP 2.0 (smart agriculture & farmer welfare)
- Lowers national food import bill
- Scalable, offline-capable, and designed for real smallholder farmers

Slogan: “Satu Gambar, Banyak Harapan untuk Ladang Anda”

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